Lab 0: Understanding MicroGPT

A Guided Conversation with AI About the Prediction Machine

ECON 470 — Microeconomics of Artificial Intelligence

Spring 2026

1 What This Is

In Lecture 1 you saw a complete GPT in 200 lines of code. That's a lot to absorb in one sitting. This lab gives you a chance to go back through the material at your own pace, with an AI tutor that has been specifically designed for this course.

You may have all your materials open: Karpathy's blog post, the lecture slides, Gans Chapters 1 and 2, the code itself. The goal is *comprehension*, not recall. But the tutor will still push you to think before it explains — because understanding something you worked for sticks better than understanding something you were handed.

This lab is ungraded. You submit a transcript for completion credit only. There are no wrong answers and no penalties. The only way to do it badly is to not engage.

2 Before You Start

Have these materials available (all were provided in class or on Canvas):

- **Karpathy's blog post:** https://karpathy.github.io/2026/02/12/microgpt/
- **The code:** https://gist.github.com/karpathy/8627fe009c40f57531cb18360106ce95
- **Lecture 1 slides**
- **Gans, Chapters 1 and 2**

You do not need to have read all of these cover to cover. Skim what you can, and let the tutor help you fill in the gaps.

3 Instructions

3.1 Step 1: Open the MicroGPT Tutor

Click the link below to open a custom AI tutor built for this assignment:

> https://chatgpt.com/g/g-69cbe9820bb481918c188ef086d7a2d8-microgpt-tutor

You will need a ChatGPT account (your Cal Poly account works). The tutor already knows the course material, the blog post, the code, and the economic framework — you do not need to set anything up.

3.2 Step 2: Start a conversation

The tutor will offer you a few ways to begin. You can:

- Start from the beginning of the pipeline
- Jump to a specific part that confused you in lecture
- Ask how the code connects to economics

Pick whichever feels right. There is no wrong starting point.

3.3 Step 3: Work through the pipeline

The tutor will walk you through the MicroGPT pipeline one stage at a time. For each stage, it will ask you what you think is going on before it helps you understand it. You can ask questions at any point — about the code, the blog post, or how it connects to Gans.

There are **six stages** to work through:

Stage	What it does	Time
1. Tokenization	Turns characters into numbers	~3 min
2. Embeddings	Gives each token a "meaning" as a vector	~5 min
3. Attention	Decides which earlier tokens matter	~7 min
4. Loss function	Measures how wrong the prediction was	~5 min
5. Training	Adjusts parameters to reduce error	~5 min
6. Inference	Generates new text	~3 min

After the six stages, the tutor will ask you a few wrap-up questions connecting the whole pipeline to economics.

Total time: About 30–40 minutes. Take more if you need it.

4 What to Submit

Share your conversation and submit the link on Canvas under the **"Understanding MicroGPT"** assignment. To get your link: click the conversation title at the top of your ChatGPT chat, select **Share**, and copy the URL.

We are looking for evidence that you engaged with the material — asked questions, tried to answer before being told, worked through things you found confusing. We are *not* grading for correctness.

5 Tips

- **If something from the lecture didn't make sense, start there.** Tell the tutor: "I didn't really understand the attention part. Can we start there?"
- **Point at specific things.** "On line 108 of the code, what does `gpt(token_id, pos_id, keys, values)` actually do?" is better than "explain the model."
- **Use your materials.** If the tutor says something and you want to check it against the blog post or the slides, do that. This is open-book by design.
- **Ask "why does this matter for economics?"** whenever a technical explanation starts to feel disconnected from the course.